

Tensor networks and molecular electronic structure

A validated study of the H_{10} chain

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Abstract. This is a methodological study. We build an end-to-end pipeline linking tensor-network electronic structure to a machine-learned energy surface, and deliberately choose a system simple enough, the linear H_{10} chain in a minimal basis, that every step remains checkable against an exact answer.

Electronic structure. Restricted Hartree–Fock, second-order Møller–Plesset perturbation theory, coupled cluster with singles and doubles and its perturbative-triples correction, full configuration interaction by Davidson diagonalisation, and SU(2)-adapted DMRG [1] with a matrix-product-operator Hamiltonian. Both correlated solvers consume a single integral tensor, so that any disagreement is attributable to the solvers rather than to their inputs. From the converged matrix product state we extract symmetry-conserved quantum-number-resolved Schmidt spectra, single-orbital entropies and two-orbital mutual information [2, 3], together with the exact decomposition of the bipartite entropy into charge and configurational parts.

Learning. A two-parameter (R, δ) surface of 77 geometries at seven levels of theory. Four models are compared at matched label budgets, splits and random seeds: a constant correction, a tensor-product B-spline, and a multilayer perceptron trained either on the correlated energy directly or on the correlation energy alone (Δ -learning, [4]). Evaluation uses repeated random splits and blocked-coordinate holdouts that force extrapolation, learning curves in the label budget, automatic-differentiation derivatives of the fitted surface, and seed-ensemble spread as an uncertainty estimate.

What the system is for. The physics here is textbook and no result is offered as new chemistry. The contribution is the protocol, and three measurements it makes falsifiable: that entanglement entropy predicts how hard a state is to represent but not the bond dimension a target **energy** accuracy needs, the conversion between the two varying by two orders of magnitude across the surface; that Δ -learning helps or hurts according to a measurable property of the target, losing by an order of magnitude where the correlation energy is the less smooth of the two; and that entanglement diagnostics do not identify where the fitted surface is untrustworthy, being beaten by the spread of a model ensemble that costs nothing.

1 Introduction

Density matrix renormalisation group methods [1] have become standard for strongly correlated molecular systems [5], and their diagnostics — orbital entropies and mutual information [2, 3, 6] — are widely used to characterise electronic structure. Separately, Δ -learning [4] is a standard device for learning correlated energies from cheap baselines. This report joins the two in one pipeline and examines what the tensor-network half actually contributes to the learning half, on a system small enough that every step can be checked against an exact answer.

Nothing here is new physics or a new method. Hydrogen chains are a long-established benchmark [7, 8]; the entanglement diagnostics are standard [2, 3]; Δ -learning is established [4]. The point of using a toy system is precisely that it removes any excuse for an unverified claim: the contribution is the protocol — matched budgets, blocked holdouts, free baselines that a proposed signal must beat — and the willingness to report what it measured when the answer was unwelcome.

1.1 What is claimed and what is not

We claim: that the energies reported are converged to the stated tolerances against two independent solvers; that the comparisons between learning models are matched in label budget, split and seed; and that the negative results are measurements rather than failures to tune.

We do not claim: chemical accuracy for the **molecule** (a minimal basis is exact for a model, not for H_{10}); transferability to other systems; any novel method; or any acquisition or active-space-selection rule.

2 Model and conventions

The system is a linear chain of ten hydrogen atoms in the STO-3G basis, with nearest-neighbour spacing R and a bond-alternation parameter δ , so that successive bond lengths are $R \pm \frac{\delta}{2}$. This gives a two-parameter family with a genuine Peierls-type distortion at $\delta \neq 0$ [8]. Restricted Hartree–Fock canonical orbitals define the active space; no orbitals are frozen, so the active space is exactly ten orbitals and ten electrons at half filling.

Conventions fixed throughout, and asserted by tests rather than by review: energies in hartree, distances in ångström, C_1 point group (block2 runs without spatial symmetry), and the same integral tensor — identified by a SHA-256 digest — handed to both solvers, so that a disagreement can never be attributed to the two codes having received different Hamiltonians.

3 Methods

3.1 Electronic structure

Seven levels of theory are computed at every geometry: restricted Hartree–Fock, MP2, CCSD, CCSD(T), full configuration interaction, and DMRG at two bond dimensions. The two correlated solvers that must agree are exact diagonalisation,

$$E_{\text{FCI}} = \min_{\mathbf{c}} \frac{\mathbf{c}^\dagger \mathbf{H} \mathbf{c}}{\mathbf{c}^\dagger \mathbf{c}}, \quad (1)$$

obtained by Davidson iteration in the full $M_S = 0$ determinant space, and SU(2)-adapted DMRG,

$$E_{\text{DMRG}}(\chi) = \min_{\psi \in \mathcal{M}_\chi} \frac{\langle \psi, \hat{H} \psi \rangle}{\langle \psi, \psi \rangle}, \quad (2)$$

minimised over matrix product states $\mathcal{M}_\chi$ of bond dimension at most χ , with the Hamiltonian represented as a matrix product operator. Working in the spin-adapted basis confines the calculation to the singlet sector by construction, where the determinant-space solver sees the whole $M_S = 0$ space including triplets.

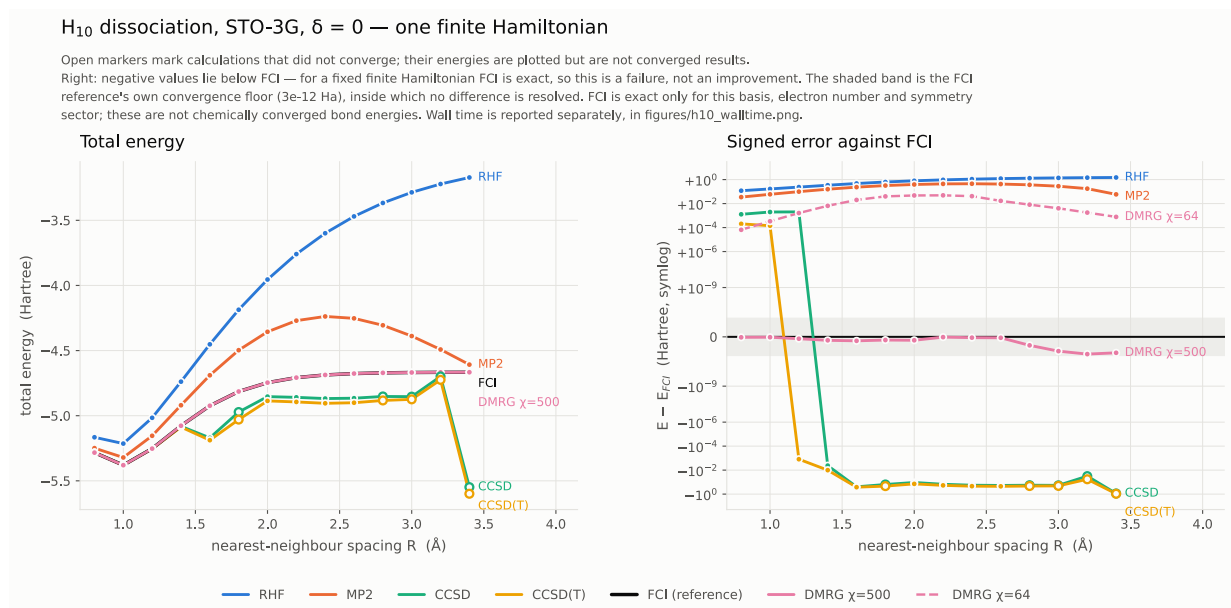


Figure 1: The dissociation curve across the method ladder (left) and the signed error against exact diagonalisation (right). The error axis is symmetric-logarithmic below the two-solver floor, which is shaded. Points where coupled cluster failed to converge are omitted rather than plotted at whatever the solver returned.

Both solvers receive one integral tensor, identified by a SHA-256 digest that is re-checked after each run — the DMRG backend writes into arrays it is handed, so the digest also guards against a solver silently perturbing the other’s input.

3.2 The N_2 active-space ladder

H_{10} in a minimal basis has no orbitals outside its active space, so DMRG there can only reproduce exact diagonalisation. N_2 in cc-pVDZ does: after freezing the two nitrogen $1s$ cores, 26 orbitals hold 10 valence electrons. Six nested active spaces are computed at three bond lengths,

$$\text{CAS}(10e, 8o) \subset \text{CAS}(10e, 12o) \subset \dots \subset \text{CAS}(10e, 26o) \subset \text{CAS}(14e, 28o), \quad (3)$$

spanning 3.1×10^3 to 1.4×10^{12} determinants. Exact diagonalisation is affordable to 16 orbitals and no further, which splits the ladder into an overlap region where both solvers run and a DMRG-only region above it, on the same molecule, basis and orbitals. $\text{CAS}(10e, 26o)$ is frozen-core FCI in this basis and $\text{CAS}(14e, 28o)$ is full CI.

Two settings are fixed by measurement rather than convention. The Davidson subspace is sized against a memory budget: the project default of 200 vectors, chosen when a CI vector was 0.5 MB, would ask for 61 GB at 16 orbitals. Halving the subspace that is used, from 28 to 16, moves the 16-orbital reference by less than 2×10^{-13} hartree, so it is converged. The self-consistent field that **defines** the orbitals runs single-threaded: on 32 threads it returns different orbitals run to run, and while the SCF energies agree to 3×10^{-13} hartree, the resulting active-space Hamiltonians differ enough to move the CASCI energy by 6.4×10^{-10} hartree. Since CASCI is not stationary with respect to orbital rotations, that noise enters the reference directly.

Orbitals are ordered along the matrix product state in two ways — by orbital energy, and by the mutual-information (Fiedler) ordering implemented in `block2` [9], which is standard practice in production DMRG codes. A permutation of orbitals cannot move an eigenvalue, so the two differ only in representational cost; that they must agree is used below as a diagnostic.

3.3 Entanglement diagnostics

From the converged matrix product state we extract the quantum-number-resolved Schmidt spectrum at every bond, the single-orbital entropy s_i , and the two-orbital mutual information [2, 3]

$$I_{ij} = \frac{1}{2}(s_i + s_j - s_{ij}), \quad I_{ii} = 0. \quad (4)$$

The bipartite von Neumann entropy decomposes exactly into a charge part and the entanglement remaining once the charge is known,

$$S(l) = S_{\text{number}}(l) + \sum_N p_{l,N} S_N(l), \quad (5)$$

which is used as a numerical invariant: its three terms come from the pooled spectrum, the charge marginal and the per-charge spectra, so none can be wrong in isolation while it holds. It also separates two effects a single entropy conflates — stretching H_{10} from $R = 1.0$ to 2.4 Å multiplies the number entropy by 3.2 but the configurational part by 8.7, as entanglement moves **inside** fixed charge sectors. These quantities are basis-dependent and are computed in both canonical and Löwdin-orthogonalised bases to make that explicit.

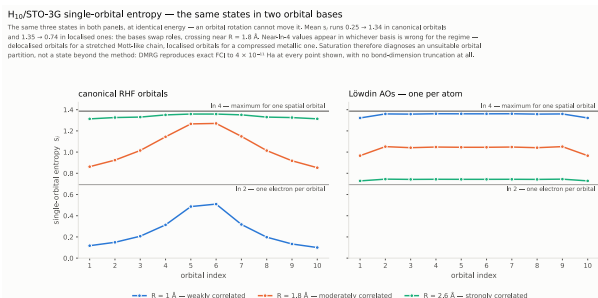


Figure 2: Single-orbital entropy.

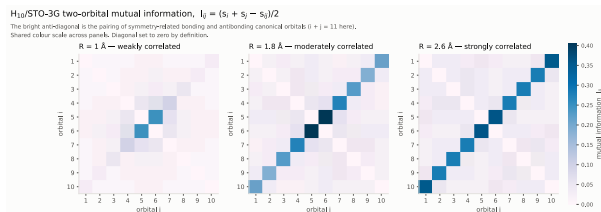


Figure 3: Two-orbital mutual information.

3.4 Learning protocol

The dataset is the (R, δ) grid of 77 geometries. Exact diagonalisation converged at all 77; coupled cluster converged at 43, and its failures cluster at stretched and positively dimerised geometries rather than scattering — which constrains the split design, since a δ -blocked holdout would load one fold with nearly all of them.

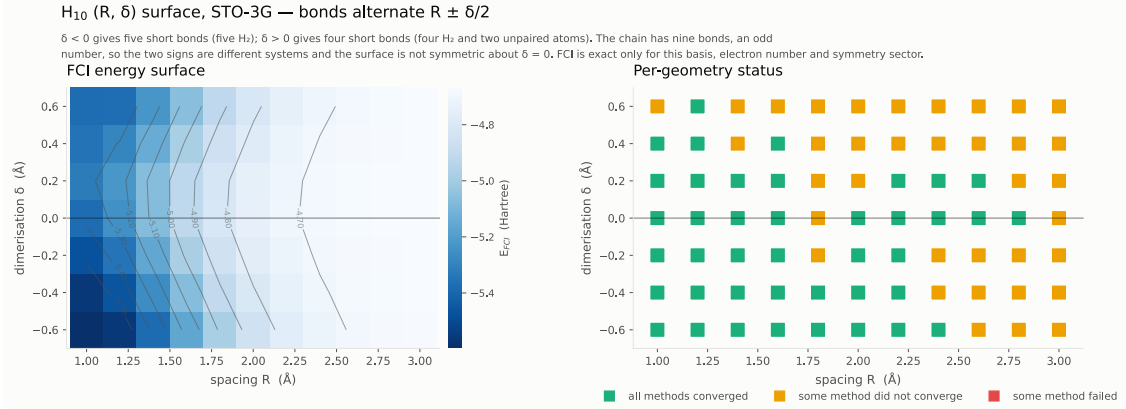


Figure 4: The (R, δ) energy surface and the per-geometry convergence status of each method. The coupled-cluster failures cluster rather than scatter, which is what constrains the split design.

Four models are compared: a constant correction, a tensor-product B-spline whose capacity is tied to the training-set size, and a multilayer perceptron trained either on E_{FCI} directly or on $E_{\text{FCI}} - E_{\text{RHF}}$ with the baseline restored at prediction time. The constant correction is Δ -learning with zero capacity and exists to make the Δ -learning claim falsifiable.

Every model at a given split and budget receives the **same** drawn training indices, so matched budgets are a property of the code rather than of discipline. Evaluation uses five repeated random splits and three blocked-coordinate holdouts in R — both ends of the range and one interior gap — so that interpolation and extrapolation are reported separately. Learning curves run over eight label budgets; three model seeds per point give an ensemble spread used later as an uncertainty estimate. Derivatives of the fitted surface are obtained by automatic differentiation and checked against central differences.

4 Numerical convergence

This section documents the convergence checks behind every number that follows. It contains no findings; it is the due diligence the rest of the report depends on, recorded because the settings are not the defaults and the reasons are worth stating.

4.1 Agreement between solvers

System	$ E_{\text{DMRG}} - E_{\text{FCI}} $	$\langle S^2 \rangle$	sum rule	Schmidt norm
N ₂ , 1.1 Å	1.7×10^{-13}	4.5×10^{-17}	2.2×10^{-13}	1.8×10^{-14}
H ₁₀ , $R = 1.0$ Å	1.1×10^{-13}	1.8×10^{-16}	2.2×10^{-16}	8.9×10^{-16}
H ₁₀ , $R = 2.4$ Å	2.5×10^{-13}	1.7×10^{-15}	8.9×10^{-16}	6.7×10^{-16}

Table 1: Agreement between the two solvers, with the spin sector, the entropy sum rule of Equation 5 and the Schmidt normalisation certified alongside the energy. Hartree or dimensionless.

Across 169 comparisons at converged bond dimension the two solvers agree to 2.3×10^{-12} hartree in the worst case and 1.8×10^{-13} in the median. That worst case is quoted throughout as the floor below which the two methods cannot be distinguished.

4.2 Solver settings

Three settings differ from the defaults of the underlying packages, each because the default was measurably insufficient on this system.

The Davidson subspace of the determinant-space solver was raised from 12 to 200. At stretched geometries the low-lying spectrum is dense — at $R = 3.4$ Å the nearest state to the singlet ground state is a **triplet** 4.6×10^{-5} hartree above it, while the nearest singlet is $2.6 \times$ further — and a 12-vector subspace restarts repeatedly and loses the directions it needs. The progression at that geometry is not monotone, so agreement between two nearby subspace sizes is not evidence of convergence:

Subspace	Error	Wall time
12 (package default), conv_tol 10^{-12}	2.52×10^{-9}	14.9 s
12, conv_tol 10^{-14}	2.42×10^{-11}	23.3 s
30	3.91×10^{-12}	7.0 s
120	3.59×10^{-12}	6.8 s
200	0 (best)	8.0 s
300	0 (best)	7.8 s

Table 2: Convergence of the determinant-space reference with Davidson subspace size at $R = 3.4$ Å. The wider subspace is also faster, converging rather than restarting. Two further knobs were measured and change nothing: the preconditioner space (400 versus 4000) and a singlet-adapted solver, which returns identical energies about 40% slower.

The DMRG local-eigsolver threshold and sweep stopping tolerance are both 10^{-14} throughout. A loose local solver is indistinguishable from bond-dimension truncation in the final energy, so leaving it at a default while tightening everything else confounds the two. This is enforced by a test: no schedule may ship a looser threshold and no script may assemble its own.

4.3 Behaviour of the bond-dimension convergence

Three properties of this system’s DMRG convergence bear on how the results below should be read.

The error moves in plateaus. Adjacent bond dimensions give energies agreeing to $\sim 10^{-13}$ over wide ranges and then jump by 10^{-4} ; at $R = 3.0$ Å the plateau extends from $\chi = 128$ to 256 and breaks only at 384. The plateaus are variational rather than an artefact of the optimiser: they survive a low- χ warm-up ramp, are unchanged at 14, 30 or 60 sweeps, and are reproduced when the optimisation is started from the exact state compressed to that bond dimension. Compressing the converged $\chi = 384$ state to $\chi = 192$ costs 1.28×10^{-2} hartree, confirming no state of that bond dimension holds the exact answer. The practical consequence is that comparing two neighbouring bond dimensions can report agreement to 10^{-13} while sitting 10^{-4} hartree away.

The discarded weight is not an error bar. Within one geometry the expected proportionality holds — at $R = 1.0$ Å the ratio of energy error to discarded weight is 0.11, 0.057, 0.055 across $\chi = 128, 192, 256$. Across geometries it does not, for reasons quantified in Section 5.1. In one case at H_{12} a run reports a discarded weight of 3.3×10^{-7} , below its own requested cutoff, while its energy is 4.2×10^{-3} hartree wrong.

Low bond dimensions are seed-dependent at stretched geometries. Three matrix-product-state initialisations agree to within $1.3 \times$ out to $R = 2.2$ Å, but spread by $10.3 \times$ at $R = 2.6$ Å and $58.7 \times$ at $R = 3.0$ Å. No low- χ tier is therefore used as a fidelity level anywhere in this work.

DMRG error against FCI versus bond dimension, H₁₀/STO-3G

Line is the median of three MPS seeds, band is their full range. Dashed line is the two-solver floor (2.3×10^{-12} Ha).

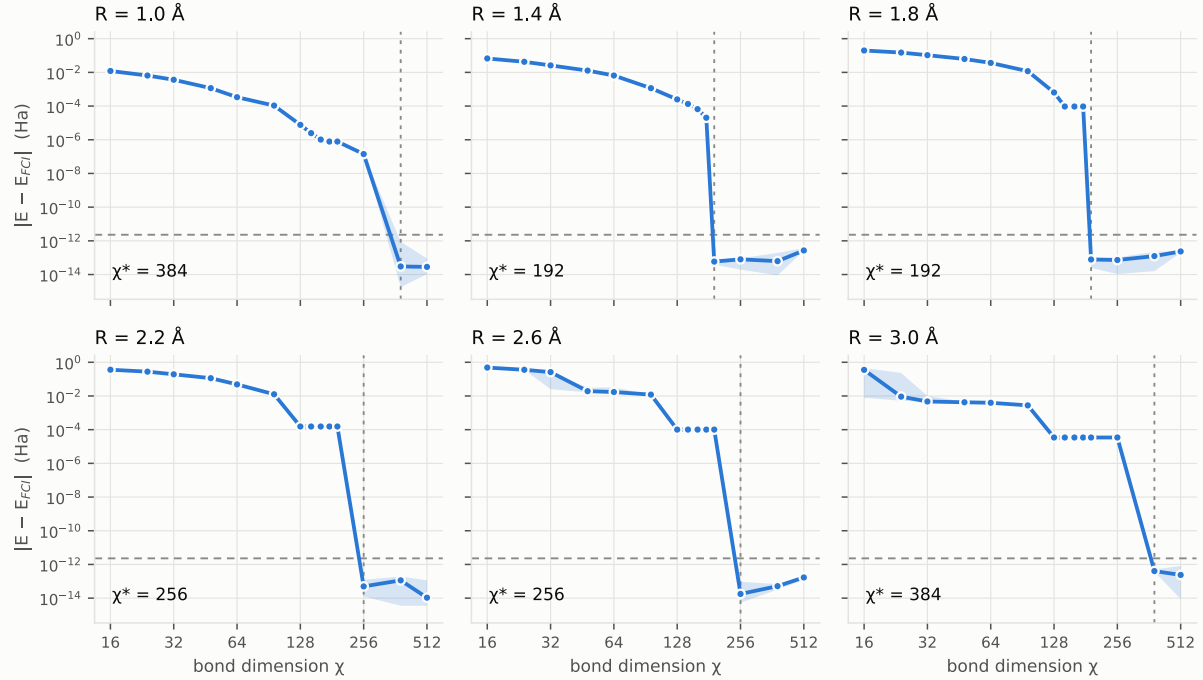


Figure 5: Error against bond dimension, one panel per spacing. Line is the median of three initialisations, band their full range, dashed line the two-solver floor.

5 Findings

5.1 Entropy predicts representational cost, not energetic cost

Let χ^* be the smallest bond dimension reaching 10^{-9} hartree against the reference. Measured this way the entropy is uninformative: χ^* is U-shaped (384, 192, 192, 256, 256, 384 across $R = 1.0$ to 3.0 Å) while the entropy rises monotonically (0.84 to 4.67), and the Pearson correlation of $\ln \chi^*$ with S is -0.08 .

That is not a failure of the standard argument, but of the question. Taken from the exact Schmidt spectrum, the number of states required for a fixed **truncation** error does follow the entropy: the correlation of $\ln n$ with S is $+0.85$ at a discarded weight of 10^{-6} and $+0.86$ at 10^{-9} . Rényi orders $\alpha = \frac{1}{2}, 2, \infty$ behave the same, so the choice of entropy measure is not the issue.

What separates the two is the conversion from discarded weight to energy, and it is not a constant of the method:

R (Å)	1.0	1.4	1.8	2.2	2.6	3.0
$\Delta E/w$	0.109	0.125	0.033	0.0032	0.0020	0.00082

Table 3: Energy error per unit of discarded weight at a common bond dimension of 128 — a monotone fall of $133 \times$ across the surface.

Discarding a given amount of Schmidt weight costs two orders of magnitude more energy at the compressed end than at the stretched end. This is consistent with the composition of the discarded tail, which at stretched geometries is dominated by near-degenerate spin configurations: they carry entropy but almost no energy. It is also the same fact as the discarded-weight caution above, seen from the other side.

The usable statement is therefore narrower than the one usually made: entropy is a good guide to how hard a state is to **represent**, and not a guide to the bond dimension a target **energy** accuracy demands, unless the energy per unit of discarded weight is known to be comparable across the systems being compared.

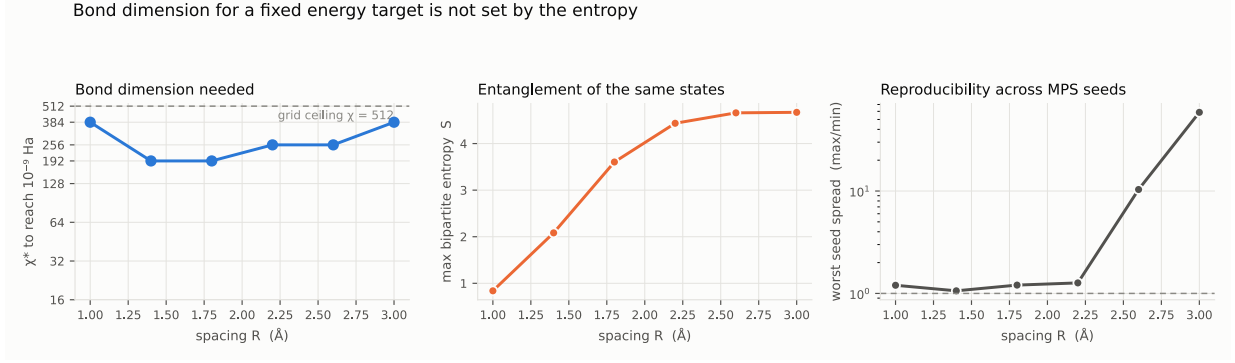


Figure 6: Bond dimension for a fixed energy target, entanglement, and reproducibility across seeds, against chain spacing.

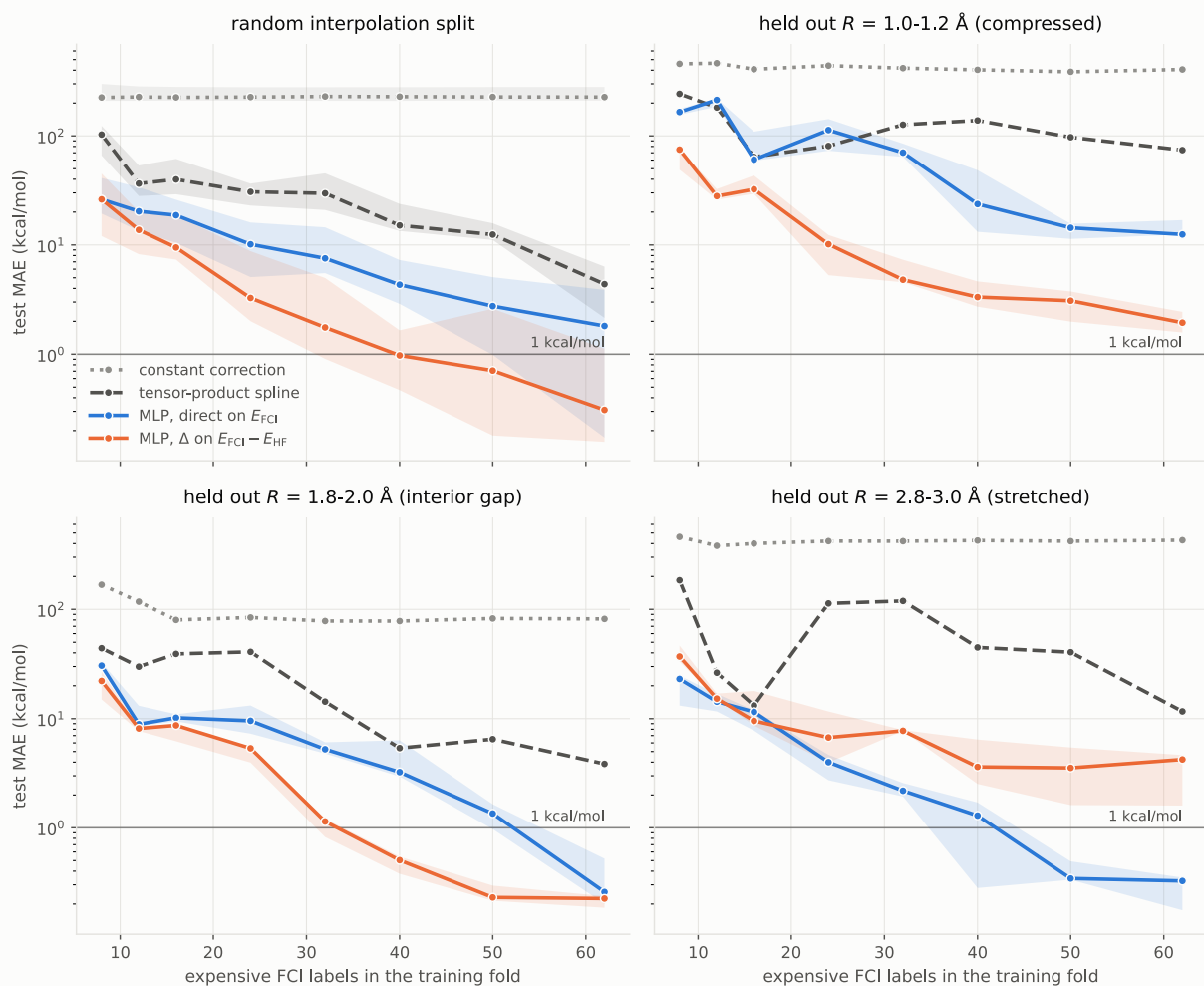
5.2 Δ -learning is conditional, and the condition is measurable

Split	direct	Δ	ratio
random (5 draws)	2.9×10^{-3}	4.9×10^{-4}	0.17
blocked, compressed	2.0×10^{-2}	3.1×10^{-3}	0.16
blocked, interior gap	4.1×10^{-4}	3.6×10^{-4}	0.87
blocked, stretched	5.2×10^{-4}	6.7×10^{-3}	13.0

Table 4: Test mean absolute error in hartree at the largest label budget.

Δ -learning wins on two splits, ties on one, and loses by an order of magnitude on the stretched block. The condition separating these is a property of the targets that can be measured in advance: the correlation energy carries about half the curvature of the total energy per unit of its own range in the compressed and middle regions, and $2.8 \times$ more in the stretched region, where the total energy flattens toward dissociation while the correlation energy is still changing rapidly. Δ -learning helps where the baseline’s error is the smoother of the two targets, and not otherwise. It is not simply a smaller target: the Δ surface spans a wider range than the total energy.

H₁₀/STO-3G (R, δ) surface: matched surrogate learning curves



Median over 5 random splits (first panel) or 3 model seeds (blocked panels); band is the full range across splits and seeds. All four models share one draw of training indices at every label count. Δ -learning wins on interpolation and into the compressed wall, and loses on the stretched block where the correlation energy is the less smooth of the two targets. Interpolation on one H₁₀ surface, not molecular transferability.

Figure 7: Learning curves at matched label budgets. The two neural models share an architecture and differ only in whether the baseline is subtracted.

5.3 Entanglement diagnostics do not identify where the surrogate fails

The surrogate consumes geometry alone, so the natural route to making the tensor network load-bearing is for an entanglement descriptor to flag untrustworthy predictions. Both surrogates were fitted at their largest budget on all eight splits with three seeds, the per-geometry held-out error recorded, and each candidate signal ranked against it. Correlations are taken **within** a split before pooling, because the errors differ by two orders of magnitude between the random and blocked splits.

Signal	direct	Δ	direct, R controlled	Δ , controlled
max bipartite entropy	-0.596	0.098	-0.287	0.146
configurational entropy	-0.598	0.096	-0.364	0.112
ensemble disagreement	0.760	0.622	0.526	0.599
distance to training set	0.340	0.460	—	—
spacing R	-0.510	0.120	—	—

Table 5: Pooled within-split Spearman correlation with the absolute held-out error over 117 points. Ensemble disagreement and the two geometric signals cost nothing; the entanglement descriptors require a converged DMRG run.

The spread of a three-seed model ensemble — already computed, therefore free — is the strongest signal for both models, before and after controlling for the spacing. About half of the apparent entanglement signal is the spacing itself, which the model already consumes. For Δ -learning, the model that actually fails, the entanglement correlation is indistinguishable from noise. The residual changes sign between the two models, so it tracks the validity of the Hartree–Fock **baseline** rather than the trustworthiness of the surrogate; a signal that reverses depending on which model it is asked about is not a trust signal.

5.4 System shape, not system size, decides whether DMRG is economical

On this system the tensor network is never the cheaper route to a label: a converged DMRG run costs 5.4 s against 2.4 s for exact diagonalisation. The standard expectation is that this reverses with size, since the determinant count grows combinatorially while the bond dimension needed for fixed accuracy need not.

Exact diagonalisation does run out first, as arithmetic requires: one Davidson vector is 1.3 GB at H_{16} and 18.9 GB at H_{18} . But at H_{12} and $R = 3.0 \text{ \AA}$, against a reference costing 44 s, DMRG plateaus at $2.5\text{--}4.2 \times 10^{-3}$ hartree and stays there, unchanged at 14, 30 or 60 sweeps and 366 s at a bond cap of 512. The surrogate of Table 4 has a test error of 4×10^{-4} hartree, so such labels would be an order of magnitude worse than the model trained on them — and beyond H_{16} there is no reference against which to notice.

The reason is the shape of the system rather than its size. H_n in a minimal basis has exactly n orbitals and n electrons: half filling, maximal entanglement per site, and no weakly correlated orbitals to compress. Production DMRG wins in the opposite regime, with many orbitals carrying structured entanglement [10, 11]. Lengthening the chain moves along the unfavourable shape rather than out of it, which is a design lesson for choosing a benchmark system rather than a property of the method.

A related correction is recorded. An earlier form of this argument held that stretched chains are gapped one-dimensional systems obeying an area law, so the bond dimension should saturate. That is false for the **uniform** chain, which approaches a Heisenberg antiferromagnet and is critical, its entanglement growing logarithmically in length; the gap in this system class comes from dimerisation. Consistent with that, at $R = 3.0 \text{ \AA}$ a dimerisation of $\delta = 0.4 \text{ \AA}$ lowers the maximum bipartite entropy from 4.671 to 3.970 at H_{10} . Whether it saturates with length could not be established, because the longer dimerised chains did not converge well enough to tell.

5.5 Above the exact wall, the only working convergence check was two orbital orderings

Below 16 orbitals, DMRG reproduces exact diagonalisation in the same active space and the cost of doing so depends strongly on orbital ordering. Writing χ^* for the smallest bond dimension within 10^{-9} hartree of the exact answer, the mutual-information ordering needs a quarter of the bond dimension at 12 orbitals, and at 16 the orbital-energy ordering does not reach that accuracy anywhere on the grid tried.

Above 16 orbitals there is no reference, and the protocol has to carry the claim. What that protocol is worth was established by a failure. With the mutual-information ordering at 16 orbitals, an earlier sweep schedule converged to a state 0.19 hartree above the ground state — at every bond dimension and both seeds at $1.6 \text{ \AA}$ — while reporting a final discarded weight of 5×10^{-10} . At $2.4 \text{ \AA}$, raising χ from 500 to 1000 at fixed seed moved the answer away from the correct one. Three controls isolated the cause: exact diagonalisation of the permuted integrals reproduced the canonical energy to 2.8×10^{-14} hartree, so the Hamiltonian was right; a flat schedule at the same seed and χ converged correctly, so the ordering was not to blame; and a schedule whose perturbative noise extended into the full- χ sweeps also converged. The defective ramp had spent its entire noise budget while χ was a sixteenth of its target, reaching full bond dimension with no mechanism left to leave a local minimum.

The diagnostics available without a reference all certified that state: discarded weight 5×10^{-10} , seeds agreeing to 10^{-12} , and an entanglement entropy lower than the correct state’s at one geometry and higher at another. Only the disagreement between the two orbital orderings — 0.19 hartree — revealed it. This is a sharper form of the discarded-weight caution of Section 5.1: here that quantity was wrong by seven orders of magnitude. Uncertainties on the upper rungs are therefore taken as the largest of the final χ step, the ordering disagreement, and the residual sweep drift, and the ordering term dominates at every point.

5.6 The convergence protocol tested against published near-exact energies

The protocol of Section 5.5 produces an uncertainty precisely where nothing in this work can check it. Three published near-exact energies for N_2 in cc-pVDZ now test it. Coordinate-descent full CI [12] and the DMRG binding curve of Chan, Kállay and Gauss [13] give the all-electron value at $2.118a_0$ and agree with each other to 1.6×10^{-5} hartree — 37 to 73 times finer than the uncertainties under test, and 39 times finer at the geometry actually run here, which makes it a test rather than a coincidence. Semistochastic heat-bath CI [14] gives a frozen-core value at $1.0977 \text{ \AA}$.

The Hamiltonian was matched before the energies were compared. RHF at $2.118a_0$ through the same path used throughout reproduces the published -108.9493779 hartree to 2.1×10^{-8} , that value’s own rounding, which fixes basis, geometry, unit and the spherical-against-Cartesian d convention at once — cc-pVDZ with Cartesian d functions gives 30 orbitals rather than 28, converges normally, and is indistinguishable in a log file.

Two obstacles were handled rather than assumed away. The sources quote different units, 2.118 Bohr and $1.0977 \text{ \AA}$, which are two geometries $0.023 \text{ \AA}$ apart and not one value written twice. And neither is on this work’s grid, so the all-electron comparison was made by running the ladder at the published bond length — eight DMRG runs, 16.4 hours of wall time — while the published curve, tabulated on a $0.05a_0$ grid near equilibrium, interpolates onto two geometries already computed. The interpolation error is measured by leave-one-out prediction of each interior published point rather than assumed: 10^{-5} hartree near both geometries used, 6.2×10^{-5} at worst across the curve, and conservative by construction because removing a point doubles the local spacing. At $2.4 \text{ \AA}$, $4.535a_0$, the curve has ended and extrapolation is refused; an extrapolated reference would be this work’s own smoothness assumption carrying someone else’s citation.

Geometry	via	E_{DMRG}	published	error	σ	$\sigma/ \text{err} $
$2.118a_0$	run here	-109.281909	-109.282173	$+2.64 \times 10^{-4}$	7.9×10^{-4}	2.98
$1.098 \text{ \AA}$	curve	-109.280658	-109.280913	$+2.54 \times 10^{-4}$	7.3×10^{-4}	2.87
$1.600 \text{ \AA}$	curve	-109.083789	-109.084270	$+4.81 \times 10^{-4}$	1.5×10^{-3}	3.02
$2.400 \text{ \AA}$	—	-108.965255	—	—	1.4×10^{-3}	—

Table 6: The CAS(14e, 28o) rung against published near-exact energies. Hartree. Every comparison falls inside the quoted uncertainty and every DMRG energy lies above the reference, as a variational bound must.

The uncertainty is honest and loose by a consistent factor of three: the ratio of the quoted band to the error actually made is 2.87, 2.98 and 3.02 across three geometries and two independent routes to the reference, and the χ step rather than the ordering term supplies that margin in all three.

The ordering gap taken alone tells a sharper and less comfortable story. It is 1.05, 1.03 and 0.68 times the true error — at two geometries predicting it to within five percent, and at $1.600 \text{ \AA}$ **under**-estimating it by half again. The diagnostic weighted most heavily in Section 5.5 because it caught a real failure is therefore also the best-calibrated estimate of the error’s size, and for that reason the most tempting quantity to quote as an error bar; these numbers say that would be wrong. Taking the maximum over the four terms is what makes the band a bound rather than an estimate.

The frozen-core rung agrees with heat-bath CI to 1.9×10^{-5} hartree, but across a $0.0003 \text{ \AA}$ geometry offset worth 3.5×10^{-5} by CCSD(T). The agreement is smaller than the offset separating the two calculations, so the supportable statement is that the rung agrees to better than that offset. That value also carries a stochastic perturbative correction and is not variational, so it is not used as a bound anywhere.

5.7 A perturbative correction, measured against the variational answer

The reduced N_2 vignette computes NEVPT2 on CASSCF(10e, 8o), a perturbative estimate of the correlation lying outside that space. The 28-orbital rung measures the same quantity variationally. The comparison is made against CAS(14e, 28o) rather than the frozen-core rung because NEVPT2 as implemented excites out of the CASSCF core and so contains core correlation the frozen-core space does not — a difference of 3–4 mhartree, larger than the agreement under discussion. That correction is confirmed two ways: directly as $E(14e, 28o) - E(10e, 26o)$, giving -3.78 , -3.07 and -3.09 mhartree, and independently from frozen-core against all-electron CCSD(T), giving -3.89 and -3.15 mhartree where CCSD(T) still converges.

R (Å)	NEVPT2	variational	NEVPT2 — full CI	kcal/mol
1.098	-0.146323	-0.174230	+0.031684	19.9
1.600	-0.155883	-0.184996	+0.032184	20.2
2.400	-0.150373	-0.183127	+0.035839	22.5

Table 7: Correlation outside CAS(10e, 8o) in hartree: the NEVPT2 estimate, the variational measurement from frozen-core FCI, and the residual against full CI in the same basis.

NEVPT2 recovers 81–83% of that correlation and misses about 20 kcal/mol at every geometry. Whether the error worsens toward dissociation is only partly resolved: from 1.098 to 1.600 Å it moves by 0.5 mhartree against an uncertainty of 1.5 mhartree, so those two are indistinguishable, while the rise at 2.4 Å is about 2.5 times the uncertainty there. The supportable statement is that the error is flat from equilibrium to 1.6 Å and rises modestly at 2.4 Å.

The same reference makes CCSD(T)’s failure quantitative for the first time here: it lies +1.5 mhartree above full CI at equilibrium, +7.4 at 1.6 Å and +265 at 2.4 Å.

6 Limitations

A minimal basis is exact for a model, not for the molecule: none of these energies is chemically converged, and no basis-set extrapolation is attempted. The N_2 results are cc-pVDZ throughout and are equally statements about correlation treatment at a fixed basis rather than about the molecule. The upper rungs of that ladder rest on the convergence protocol of Section 5.5, which bounds the spread of what was tried rather than the true error; that protocol is tested against published near-exact energies in Section 5.6 and holds at three geometries, over-covering by a factor of three. The test does not reach the hard case: 2.4 Å is the most strongly multireference geometry here and the one where the protocol is likeliest to fail, and it lies outside the published curve, so the agreement elsewhere does not transfer to it. The references themselves are near-exact rather than exact. The (R, δ) surface is a two-parameter slice of the nine independent bond lengths of an H_{10} chain. The learning study interpolates one surface of one system; nothing here demonstrates transferability. Derivatives reported are generalised-coordinate derivatives of a **fitted** surface, not Cartesian atomic forces, and no model was trained on force labels. The entanglement diagnostics depend on the orbital basis and are not invariants of the state. No model in this work was trained on tensor-network data: the surrogates take two geometry coordinates as input and exact-diagonalisation energies as labels, DMRG-derived quantities were tested only as trust signals and reported as a negative result, and the N_2 ladder has no learning component.

7 Reproducibility

Every calculation writes a structured record carrying geometry, units, method, basis, convergence status, energies, wall time, hardware identifier, package versions, and a SHA-256 digest of the integral tensor actually consumed. Figures are generated only from stored data; the figure scripts perform no electronic-structure calculation, and this is asserted by comparing checksums of the data directory before and after. The solver-convergence conventions are themselves under test: a regression test asserts that no sweep schedule ships a loose local-solver threshold, that no script assembles its own, and that the exact-diagonalisation subspace does not fall below the value established in Table 2. Software: PySCF [15] and block2 [9].

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